GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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Abstract. We determine the joint acquisition–memory frontier of prepared causal experiments whose current task response is regenerated by a chosen acquisition action. The response laws may depend on the action, have different supports, and contain singular or atomic components. An actual occupation profile, rather than an ambient parameter dimension, characterizes both arbitrary-history checkpoint procedures and autonomous finite-state observers. Every feasible terminal occupation is realized by one live state and the prescribed output labels; no additional clock or retained random seed is used. A common mass–cost dual and a two-action reduction make the acquisition optimization explicit. Conditional domination extends the matching bounds to correlated, controlled and model-indexed experiments, including classical polynomial instruments and noncommuting quantum instruments. The regenerative frontier has a linear same-budget perturbation modulus, in contrast with the square-root modulus needed for general stopped experiments. An atomic–continuous acquired law exhibits a uniform mass-dependent memory transition. Beyond the regenerative class, we prove an exact all-randomized-observer two-state Gaussian frontier and give certified checkpoint constructions for the same stopped physical experiment. Predictive realization, finite precision, causal deficiency and all persistent and temporary resources are distinguished throughout.

1. Introduction

A finite observer does not receive a geometric object to be compressed. It executes an experiment: it chooses an action, pays for a report, retains a label, and eventually commits to a prediction before a protected test is performed. The distribution to be approximated is created by those choices. Even when the current test response is exactly visible, a stopping rule changes its law, and the cost of obtaining a favorable response cannot be removed by conditioning.

We study the passage

$$\text{prepared kernels} \longrightarrow \text{executable tests} \longrightarrow \text{predictive realization}$$
$$\longrightarrow \text{actual acquisition and terminal mass} \longrightarrow \text{finite-resource risk.}$$

The principal theorem identifies a class on which this passage is exact. Each acquisition action has a prescribed current-response law, but different actions need not have the same law, support, cost, or local dimension. We call this property *task regeneration*; it is a condition on a specified executable test, not an assertion that all hidden physical variables are independent or reset. Theorem 3.1 converts every history-dependent acquisition and stopping procedure into a scalar action-occupation profile, and conversely implements every feasible profile with one live state. Optimizing a codebook and optimizing the acquired law are therefore one constrained problem. The full causal memory price is exactly one additional live label.

Two issues make the result different from static quantization. First, action counts and selected terminal submeasures must be compatible. The selected mass belonging to an action cannot exceed its expected acquisition count times its one-call law. Second, the same codebook must serve the selected mass from every action. These conditions lead to a single mass–cost dual, in which the

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same multiplier is tested against all actions. Optimizing each action with a separate codebook and then taking an average is not the problem considered here. A finite-dimensional extremal argument reduces the executing action mixture to two actions under one scalar budget. This support reduction is a standard linear-programming mechanism; its role here is to preserve the exact acquired law and the total observer-state count.

The exact frontier is complemented by three quantitative results. Uniform conditional domination, imposed at every surviving history before selecting a program, gives matching profile bounds for correlated controlled experiments. A raw perturbation of regenerative response laws can be repaired at the *same* expected acquisition and state budgets with a linear error modulus. Finally, a mixed atomic–regular acquired law has a continuous-mass factor that vanishes at an explicit acquisition threshold. Its rate changes with the number of output labels, even though the physical carrier is fixed.

Theorem 7.1 supplies an informative-filter consequence outside task regeneration. For a symmetric positive two-state hidden chain with Gaussian reports, it characterizes the optimum among *all* Borel randomized two-state autonomous observers. The proof combines the actual pre-report occupation equations with the classical likelihood-ratio ordering. A one-parameter threshold determines both the exact mean acquisition and the attained prediction risk. A separately certified construction compares this optimum with checkpoint procedures for the same physical stopping experiment; it is not described as the unknown global checkpoint optimum.

The foundations and realizations remain distinct. Theorem 3.1 is proved before the classical and quantum kernels are introduced. Those kernels verify its conditional-domination hypotheses directly. The Gaussian theorem uses the more general occupation representation, not a false assertion that a Gaussian posterior is recovered from its last report. Full proofs of the general common-program occupation criterion, the informative Bayesian three-defect identity, global-cover stability bounds, and earlier singular and finite-budget results are retained in the technical companion [1, 2]. The present theorem does not replace them by a narrower definition.

Relation to earlier mathematics. Population trimming and optimal quantization already optimize a density bounded by a reference measure [3]. Budgeted rejection sampling optimizes acceptance under a sampling constraint [4]; recent finite-sample selection results address a different fixed-sample architecture [5]. Stationary policy reductions and finite mixtures under occupation constraints belong to the established theory of constrained Markov decision processes [6, 7]. We use these mechanisms with explicit attribution and give direct proofs for the present setting. Our claim is not that occupation measures, rejection sampling, convex duality, or likelihood-ratio tests are new. The contributions are their exact conjunction with action-dependent acquired geometry, the one-live-state implementation of the entire attainable terminal profile, the same-budget perturbation repair, and the stated singular and informative-filter consequences.

Predictive equivalence and statistical experiment comparison have a separate classical history [8, 9]. Belief compression [10], Bayes-risk quantization [11], and finite-memory filtering approximation [12] remain relevant comparisons. The present finite-state count is not a mutual-information rate, a number of stored floating-point coordinates, or a bound on temporary computation. No claim of a complete classification of causal compressibility or of a universal learning rate for unknown kernels is made.

2. Prepared experiments and the paid interface

All physical, action and report spaces are standard Borel. A prepared experiment consists of a preparation law and jointly measurable positive probability kernels

$$K_a(dx', dy \mid x), \qquad a \in \mathcal{A},$$

where the nonempty action set $\mathcal{A}$ is finite. Positivity means nonnegative measures; a uniform positive likelihood is an additional condition only when explicitly stated. After each call an observer may stop or continue. A protected audit $Z \in \mathbb{R}^d$ is performed *after* the prediction is committed. Its law and physical evolution are fixed by the experiment, not by the decoder.

Write $\mathcal{F}_{t-1}$ for the complete past, including previous private randomizations, before action A_t is chosen. The current report is Y_t. The present main class has known Borel functions f_a and compact sets $K_a \subset \mathbb{R}^d$ such that, at every legal stopped history,

$$X_t = f_{A_t}(Y_t), \qquad \mathbb{E}[Z \mid \mathcal{F}_{t-1}, A_t, Y_t, \text{current stop coins}] = X_t, \qquad \mathbb{E}[\|Z - X_t\|^2 \mid \cdots] = B. \tag{2.1}$$

The constant $B \geq 0$ is common to actions and histories. The audit is bounded. The dots denote the same pre-audit information, not a future audit result. Thus every decoder r_J satisfies

$$\mathbb{E}\|Z - r_J\|^2 = B + \mathbb{E}\|X_\tau - r_J\|^2. \tag{2.2}$$

Let $K = \bigcup_a K_a$ and $H = \operatorname{conv} K$; H is compact. Decoder values may be projected to H without increasing the excess in (2.2).

Definition 2.1 (Regenerative task and conditional envelopes)**.** A task is *regenerative* if probability measures ν_a on K_a satisfy

$$\operatorname{Law}(X_t \mid \mathcal{F}_{t-1}, A_t = a) = \nu_a \tag{2.3}$$

for every surviving history, every time and every legal choice of action. These measures are actual one-call laws under the preparation. They are not a formal reachable-set measure.

The *conditional-envelope class* replaces (2.3) by

$$\underline{b}_a \nu_a \leq \operatorname{Law}_\theta(X_t \mid \mathcal{F}_{t-1}, A_t = a) \leq \overline{b}_a \nu_a, \qquad 0 < \underline{b}_a \leq 1 \leq \overline{b}_a < \infty. \tag{2.4}$$

The fixed parameter θ is unknown to the observer. Bounds hold for every θ and every legal history outside its null set, before optimizing the program. Each ν_a must be exhibited as an actual one-call law in a specified prepared reference member. Known maps, bounds and reference laws may enter a common program; the true θ may not. The full future predictive quotient need not be the current task vector X_t; Appendix A separates them.

A checkpoint procedure may use the whole history to choose actions and a randomized stopping time $1 \leq \tau < \infty$, then encode that history once into at most m output labels. An autonomous observer has disjoint finite live and terminal sets S, J, a fixed initial live state, an action kernel $p(a \mid s)$, and a routing kernel

$$\gamma(s' \text{ or } j \mid s, a, y).$$

Every choice uses only the current live state, current report and fresh independent coins. A retained seed, an external phase counter, and rereading earlier reports are disallowed. Its entire state budget is $|S| + |J| \leq M$, including all stopping and output labels. The protected audit does not update this state.

The primary acquisition budget is $\mathbb{E}\tau \leq N$, $N \geq 1$. We also state a scalar priced version with fixed known costs $c_a > 0$ and

$$\mathbb{E}\sum_{t=1}^{\tau} c_{A_t} \leq N. \tag{2.5}$$

Here N has cost units, not automatically raw-call units; the raw count is recovered only by $c_a = 1$. Preparation and audit costs are listed separately in Section 8. Risk infima are denoted by $\mathcal{R}_{\mathrm{cp}}(N, m; c)$ and $\mathcal{R}_{\mathrm{aut}}(N, M; c)$. In (2.4) these are infima of the worst-model risk over common legal programs with a worst-model budget. Fresh output randomization cannot improve a squared loss, so fixed decoder values suffice.

The legal class is empty for an autonomous observer with $M < 2$, since a call and a subsequent stop require a live and a terminal label. No prescribed deterministic horizon is imposed. In

particular, the fixed-horizon clock deduction $M-n$ from informative filtering is not applicable to this mean-budget stopping interface.

3. The acquired action-occupation theorem

For $C=(r_1,\dots,r_m)\in H^m$ put $D_C(x)=\min_j\|x-r_j\|^2$. Define

$$\begin{aligned}\Psi_m(N;c)=\min\ &\sum_{a\in\mathcal{A}}\int D_C(x)h_a(x)\nu_a(dx), \\ &C\in H^m,\quad x_a\ge 0,\quad 0\le h_a\le x_a\quad \nu_a\text{-a.e.}, \\ &\sum_a\int h_a\,d\nu_a=1,\qquad \sum_a c_a x_a\le N.\end{aligned}\tag{3.1}$$

An infeasible minimum is $+\infty$. Here x_a is an expected acquisition count, and $h_a\nu_a$ is an *unnormalized* terminal submeasure. Repeated letters x in the integration variable and x_a in the counts have different types. We write $\Psi_m(N)$ when all costs are one.

Theorem 3.1 (Causal acquired-geometry and acquisition transfer)**.** *Let the experiment and task satisfy Section 2, and let $m\ge 1$, $M\ge 2$.*

(i) *Under task regeneration, for every real $N\ge\min_a c_a$ the minima exist and*

$$\mathcal{R}_{\mathrm{cp}}(N,m;c)-B=\Psi_m(N;c),\qquad \mathcal{R}_{\mathrm{aut}}(N,M;c)-B=\Psi_{M-1}(N;c).\tag{3.2}$$

Every feasible $(x_a,h_a)_a$ in (3.1) is realized, with its exact terminal submeasures and costs, by a homogeneous observer with one live label and at most m terminal labels. An optimal observer can be chosen to use at most two acquisition actions.

(ii) *For fixed centers the profile has the dual representation*

$$V_C(N;c)=\sup_{\lambda\ge 0,\ z\in\mathbb{R}}\left\{z-\lambda N:\int(z-D_C)_+\,d\nu_a\le\lambda c_a\quad \textit{for every } a\right\},\qquad \Psi_m=\min_{C\in H^m}V_C.\tag{3.3}$$

For $N>\min_a c_a$ the dual is attained. At the endpoint its supremum remains exact, without an assertion of attainment. The same pair (z,λ) is used for every action. No exchange of the center minimum and the dual supremum is asserted.

(iii) *Under the conditional envelopes (2.4), let $\kappa=\max_a\bar{b}_a/\underline{b}_a$. Then the common-program optima obey*

$$\Psi_m(N;c/\bar{b})\le\mathcal{R}_{\mathrm{cp}}(N,m;c)-B,\qquad \mathcal{R}_{\mathrm{cp}}(N,m;c)-B\le\kappa\Psi_m(N;c/\underline{b}),\tag{3.4}$$

$$\Psi_{M-1}(N;c/\bar{b})\le\mathcal{R}_{\mathrm{aut}}(N,M;c)-B,\qquad \mathcal{R}_{\mathrm{aut}}(N,M;c)-B\le\kappa\Psi_{M-1}(N;c/\underline{b}).\tag{3.5}$$

The right-hand bounds assert an implementation only when the respective profile is feasible. Its program is parameter independent and still has one live label.

(iv) *Consider two known regenerative response families on the same compact task space, with unit acquisition costs and the same protected audit mean and constant variance. If $\max_a\|\nu_a-\nu'_a\|_{\mathrm{TV}}\le\eta$, and $0\le D_C\le L$ for all projected decoders, then*

$$|\Psi_m(N)-\Psi'_m(N)|\le\min\{L,2LN\eta\}\qquad (N\ge 1).\tag{3.6}$$

The comparison transfers an optimal one-live-state program to a program with the same *state and mean budgets. It is not an inference that total variation controls arbitrary stopping means.*

The theorem optimizes over all nonanticipating acquisition strategies, not merely over a common fixed controller. Its exact part is also a sufficiency statement: the scalar mass inequalities characterize the entire attainable selected law on this class. Task regeneration is essential for this characterization. Under (2.4) the theorem asserts the displayed matching comparisons, not equality of selected laws. The Gaussian result of Section 7 shows how an informative experiment outside this exact class is instead handled by the raw occupation equations.

Corollary 3.2 (Regular acquired supports)**.** *Suppose each ν_a has support K_a of diameter at most a fixed constant and, for fixed $c_0, C_0, r_0 > 0$,*

$$c_0 r^{s_a} \le \nu_a(B(x,r)) \le C_0 r^{s_a} \quad (x \in K_a,\ 0 < r \le r_0), \qquad 0 < s_- \le s_a \le s_+ < \infty. \tag{3.7}$$

For unit costs and uniformly bounded envelope constants, for every $N \ge 1$,

$$\mathcal{R}_{\mathrm{cp}}(N,m) - B \asymp \min_a (Nm)^{-2/s_a}, \tag{3.8}$$

$$\mathcal{R}_{\mathrm{aut}}(N,M) - B \asymp \min_a \{N(M-1)\}^{-2/s_a}. \tag{3.9}$$

Constants depend only on the displayed regularity, diameter and envelope data and the finite action family, not on N, m, M. These exponents belong to actual one-call supports. They are not the dimension of a chosen physical or parameter carrier.

For nonregular or mixed laws it is (3.1), not a single exponent, that remains the conclusion. The exact profile allows actions to have different atomic masses and different scales; mixtures may be useful at finite budgets even when one action eventually has the smallest regular exponent. No optimal-control theorem is inferred for actions outside the declared legal family.

4. Proof of the central theorem

Lemma 4.1 (Actual action mass)**.** *For a regenerative task and any admissible history procedure, set*

$$x_a = \mathbb{E}\sum_{t=1}^{\tau} \mathbf{1}_{\{A_t = a\}}, \qquad \mu_a(D) = \mathbb{P}(A_\tau = a, X_\tau \in D).$$

Then $\mu_a \le x_a \nu_a$, $\sum_a \mu_a(K_a) = 1$, and $\sum_a c_a x_a$ is the actual expected cost. Under (2.4)*, $\mu_{\theta,a} \le \overline{b}_a x_{\theta,a} \nu_a$.*

Proof. The event $\{\tau \ge t\}$ is measurable before the tth call. Condition also on its selected action, and use (2.3) to obtain

$$\mathbb{E}\sum_{t=1}^{\tau} \mathbf{1}_{\{A_t = a, X_t \in D\}} = \nu_a(D)\mathbb{E}\sum_{t=1}^{\tau} \mathbf{1}_{\{A_t = a\}}.$$

Tonelli applies to these nonnegative sums. The terminal count is bounded by the acquired count, which proves domination. Positive minimum cost and finite cost imply $\mathbb{E}\tau < \infty$, hence terminal mass one. The same calculation with the upper envelope proves the last assertion. No law after a rejection has been conditioned back to its original value. □

Lemma 4.2 (Exact causal realization)**.** *Every feasible point of* (3.1) *has a one-live-state realization with exact action counts and exact action-indexed terminal submeasures.*

Proof. Let $T = \sum_a x_a \ge \sum_a \int h_a = 1$. At the single live label choose action a with probability x_a/T. If $x_a > 0$, after response z stop with probability $h_a(z)/x_a$, and route to a nearest center with a fixed Borel tie rule. Otherwise continue in the same live label. An action of count zero is never chosen.

At every surviving call the stopping probability is exactly $1/T$. Its action-indexed accepted submeasure is $h_a\nu_a/T$. Thus τ is geometric of mean T, the expected number of action-a calls is x_a, and the terminal submeasure is $h_a\nu_a$. The expected cost is exactly $\sum_a c_a x_a$. No random value persists after a rejection; fresh action and routing coins are used. Only one live and m terminal labels are present. This also implements feasible points with positive counts but zero accepted mass for some action. □

Lemma 4.3 (Existence and the mass–cost dual)**.** *The primal and dual statements in Theorem 3.1*(i)–(ii) *hold.*

Proof. The counts lie in the compact box $0 \le x_a \le N/c_a$. The constraints on h_a are weak-star closed in the corresponding bounded subsets of $L^\infty(\nu_a)$. For example, $h_a \le x_a$ is equivalent to all nonnegative L^1 integral inequalities and is preserved when the scalar x_a converges. The centers lie in compact H^m. Their squared distance functions converge uniformly as the centers converge, so the objective is jointly continuous for these topologies. A minimum follows. Feasibility is equivalent to $N \ge \min c_a$: necessity uses $\sum x_a \ge 1$, and sufficiency takes one cheapest action with $x = h = 1$.

For fixed C, attach multipliers $z \in \mathbb{R}$ to unit terminal mass and $\lambda \ge 0$ to the cost bound. Minimizing the Lagrangian over $0 \le h_a \le x_a$ gives

$$z - \lambda N + \sum_a x_a \left\{ \lambda c_a - \int (z - D_C)_+ \, d\nu_a \right\}.$$

Its infimum over $x_a \ge 0$ is $z - \lambda N$ precisely under the inequalities in (3.3). This proves weak duality.

For completeness, strong duality is a finite-dimensional separation argument, not a minimax exchange with the centers. Consider the convex cone of triples (terminal mass, cost, distortion) obtained from $x_a \ge 0$ and $0 \le h_a \le x_a$, with nonnegative slack in the last two coordinates. On every bounded-cost slice its generating variables are weak-star compact, so the cone is closed locally where the cost is bounded. At mass one and cost $N > c_* := \min c_a$, the lower distortion boundary has a supporting affine plane, since the projection onto (mass,cost) contains the open cone $\{u > 0, v > c_* u\}$. Its cost coefficient is nonpositive, because increasing a permitted cost cannot increase the value. Normalize the distortion coefficient to one; this is possible at an interior projected point where the finite convex value function has a subgradient. Writing the plane as distortion $\ge zu - \lambda v$ gives $\lambda \ge 0$. Testing it on all rays from one action gives exactly the dual inequalities. Equality at $(1, N, V_C)$ proves attainment.

At $N = c_*$ take $N_k \downarrow c_*$. Compactness of primal minimizers shows $V_C(N_k) \uparrow V_C(c_*)$. A dual optimizer at N_k evaluated at c_* has value $V_C(N_k) + \lambda_k(N_k - c_*) \ge V_C(N_k)$. Weak duality and passage to the limit prove the endpoint supremum formula. This argument does not require bounded endpoint multipliers. □

Proof of Theorem 3.1(i). The protected conditional projection identity (2.2) and Lemma 4.1 bound every checkpoint with m output values below by $B + \Psi_m$. An autonomous observer has at most $M - 1$ output values, since a live state is necessary. Thus it is bounded below by $B + \Psi_{M-1}$. Lemmas 4.2 and 4.3 give attaining programs with exactly the allowed architecture, proving both equalities.

We prove the asserted two-action choice directly. Start at a primal optimizer and discard an action with zero accepted mass by setting its count to zero; this cannot increase cost. Let $w_a = \int h_a \, d\nu_a > 0$ on the remaining actions. Freeze the conditional terminal law $h_a\nu_a/w_a$, the per-unit terminal cost $v_a = c_a x_a / w_a$, and its distortion $e_a = \int D_C h_a \, d\nu_a / w_a$. The original weights satisfy $\sum w_a = 1$ and $\sum v_a w_a \le N$. Minimize $\sum e_a w_a$ over this finite polytope. An extreme minimizer has at most two positive weights: with three positive weights there is a nonzero direction preserving both total mass and cost, and both sufficiently small signs remain feasible, contradicting extremality. If the cost is slack the same conclusion, or the stronger one-action conclusion, follows. Rescaling each frozen count and density by its new weight preserves $h_a \le x_a$. The new objective cannot exceed the global optimum, so it is again optimal and Lemma 4.2 realizes it. This elementary support reduction is consistent with, but does not require, general constrained-policy reduction theorems [7]. □

Proof of Theorem 3.1(iii). Fix a model and any legal observer. Lemma 4.1 gives terminal densities bounded by $\bar{b}_a x_{\theta,a}$. Put $x'_a = \bar{b}_a x_{\theta,a}$. Their cost with coefficients $c_a/\bar{b}_a$ is at most N. This proves the two lower bounds for each model and hence for the common worst-model optimum.

For the upper, choose a minimizing point for $\Psi_m(N; c/\underline{b})$. Put $T = \sum_a x_a/\underline{b}_a \ge 1$. At one live state choose a with probability $x_a/(\underline{b}_a T)$ and accept response z with probability $h_a(z)/x_a$. At every surviving history its stopping probability is at least $1/T$. Therefore $\mathbb{E}_\theta \tau \le T$. Its mean cost per

surviving call is the fixed number $\sum_a c_a x_a/(\underline{b}_a T)$, giving total mean cost at most N. Its conditional distortion contribution on a call is at most

$$\frac{1}{T}\sum_a \frac{\overline{b}_a}{\underline{b}_a}\int D_C h_a \, d\nu_a \le \frac{\kappa}{T}\Psi_m(N; c/\underline{b}).$$

Sum over the actual survival probabilities to prove the upper. The argument is uniform over the unknown fixed parameter and all conditional rows, including those after rejection. It does not claim that the accepted law equals its regenerative reference law. □

Proof of Theorem 3.1(iv). Use an optimal one-live-state program for the unprimed experiment. Its action probabilities and report routing induce a joint one-call law on the response, the continuation symbol, and terminal labels. Write $s \ge 1/N$ for the one-call stopping mass and A for the subprobability measure of accepted response–label pairs. The risk in excess of B is $\int D\, dA/s$, where D is the squared loss for that routing. The primed one-call law has accepted submeasure A' and stopping mass s', with $|s - s'| \le \eta$. Since $D\mathbf{1}_{\{\mathrm{stop}\}} \in [0, L]$, its numerator changes by at most $L\eta$.

If $s' < s$, redirect a fraction $(s-s')/(1-s')$ of continuing outcomes to one already existing terminal label. This is a fresh routing coin, costs no new label, and adds accepted mass $d = s - s' \le \eta$. If $s' \ge s$, make no change. The repaired stop mass is $s'' = \max(s, s')$, so its mean is $1/s'' \le 1/s \le N$. Its excess risk is at most

$$\frac{\int D\, dA + L\eta + L(s - s')_+}{s''} \le \frac{\int D\, dA}{s} + 2LN\eta.$$

A stop mass $s' = 1$ requires no repair. The construction is valid at $N = 1$ as well. Interchanging the two known experiments proves (3.6); the diameter bound gives the truncation at L. The repair uses the known primed one-call mass, not an unknown parameter or an unbounded-cost consequence of transcript proximity. □

Proof of Corollary 3.2. For a union of m balls of radius r about arbitrary centers, (3.7) gives ν_a-mass at most Cmr^{s_a}; if a center is outside K_a, move it to a point of intersection and double the radius. Lemma 4.1 and unit cost show that the actual terminal mass in this union is at most

$$CmN \max_a r^{s_a}.$$

Choose $r = c \min_a (Nm)^{-1/s_a}$, with c small enough for all exponents and for $r \le r_0/2$. At least half the terminal mass lies outside the balls. This proves the lower bound, with $m = M - 1$ for autonomous observers.

For the upper use only a minimizing action. We recall the required global argument. A maximal r-separated subset of an s-regular support gives a global cover with at most Cr^{-s} points, by the lower mass bound on disjoint balls. The upper mass bound, conversely, produces m separated points at distance $cm^{-1/s}$. For t above a fixed constant, their disjoint radius $C(tm)^{-1/s}$ balls have total reference mass at least $1/t$. Accept a normalized restriction to that union; its density is at most t and distortion at most $C(tm)^{-2/s}$. For bounded $t \ge 1$ use the whole-law global cover instead. These statements retain the actual mass and prove $\Psi_m^{\nu_a}(t) \asymp (tm)^{-2/s_a}$. Apply the envelope upper with $t = \underline{b}_a N$ when this is at least one. When $1 \le N < 1/\underline{b}_a$, stop after one call and use the global cover. Constants are uniform over the finite action family. No trajectorywise contraction or asymptotic dimension theorem is inserted into this construction. □

5. Singular acquired mass and a memory transition

The following result is an extension of the retained mass-constrained profile, not an assertion that all singular supports have a dimension. It keeps the physically acquired weight of a continuous component through the transition to atomic selection.

Theorem 5.1 (Atomic–regular transition)**.** *Consider one regenerative action with unit cost and actual one-call law*

$$\nu_b = (1-b)\delta_{x_0} + b\nu, \qquad 0 < b \le 1, \tag{5.1}$$

where ν is a nonatomic compact s-regular probability satisfying (3.7)*; x_0 is in a fixed bounded task domain. Put*

$$r = r(N,b) = [1 - N(1-b)]_+.$$

For $N \ge 1$ and $m \ge 2$,

$$c\frac{r^{1+2/s}}{(Nb)^{2/s}m^{2/s}} \le \Psi_m^{\nu_b}(N) \le C\frac{r^{1+2/s}}{(Nb)^{2/s}(m-1)^{2/s}}. \tag{5.2}$$

The constants are independent of N, m, b. For any finite $m \ge 1$, zero excess is possible exactly when $N(1-b) \ge 1$. In particular the complete autonomous risk is $B + \Psi_{M-1}^{\nu_b}(N)$, with no discarded rejected calls.

For $x_0 = 0$, ν uniform on $[1,2]$, $0 < b \le 1/8$, and $1 \le N < 1/(1-b)$, the single-output profile is exactly

$$\Psi_1^{\nu_b}(N) = r\left(1 + \ell + \frac{\ell^2}{3}\right) - r^2\left(1 + \frac{\ell}{2}\right)^2, \qquad \ell = \frac{r}{Nb}. \tag{5.3}$$

For this uniform law, (5.2) *holds with lower constant* $1/12$ *and upper constant* $1/12$ *as written with m and $m-1$, respectively.*

Proof. Every admissible terminal law has atomic mass at most $N(1-b)$; its continuous mass u is therefore at least r. If $u > 0$, its normalized continuous part has density at most Nb/u with respect to ν. The regular profile estimate from the proof of Corollary 3.2 gives a distortion at least

$$u\,c\left(\frac{Nb}{u}m\right)^{-2/s} \ge c\frac{r^{1+2/s}}{(Nb)^{2/s}m^{2/s}}.$$

The argument applies to arbitrary centers, including a center at the atom. For the upper when $r > 0$, put mass $1 - r = N(1-b)$ at the atom, use one center there, and use an optimal $(m-1)$-center selected regular law of total mass r and density cap Nb. This is admissible because $Nb/r \ge 1$. Its distortion gives the upper. When $r = 0$, accept only the atom; the mean cost is $1/(1-b) \le N$. Conversely, zero distortion with finitely many centers forces the terminal probability onto atoms, and the only atom has available mass at most $N(1-b)$. Compact attainment from Lemma 4.3 rules out a zero infimum without a zero minimum. This proves the exact threshold.

For (5.3), any feasible selected law has continuous mass at most $Nb < 1/7$. Its best center, its mean, is therefore below $2/7 < 1/2$. At that center the atom is strictly closer than every point of $[1,2]$. The elementary bounded-density threshold argument then fills the atom to its capacity and chooses the closest remaining interval, $[1, 1+\ell]$, with density Nb. To spell out that argument, for a fixed center and threshold ζ, the integral of $(D_C - \zeta)(\widetilde{h} - h)$ is nonnegative whenever h fills the strict sublevel and partially fills its boundary; equal mass cancels ζ. Thus a jointly optimal center and density has exactly this form. Its continuous mass is r, its first moment is $r(1 + \ell/2)$, and its second moment is $r(1 + \ell + \ell^2/3)$. Subtracting the squared first moment proves the formula.

For the asserted explicit constants, a measure on the line with mass u and Lebesgue density at most H has, for any m centers, distortion at least $u^3/(12H^2m^2)$. Indeed, split it into nearest-center cells of masses u_j; filling the interval of length u_j/H symmetrically about its center minimizes each moment and gives $u_j^3/(12H^2)$. Jensen's inequality gives $\sum u_j^3 \ge u^3/m^2$. Here $H = Nb$ and $u \ge r$. For the upper put $m-1$ disjoint intervals of length $r/[Nb(m-1)]$ inside $[1,2]$, centered at their midpoints, and fill them at density Nb; their total length is at most one. Add the atomic mass. This attains the stated upper constant. □

For a fixed positive small b, as $N \uparrow 1/(1-b)$, (5.3) is of order r, whereas two or more output labels allow order r^3 in the one-dimensional example. These are two distinct finite-memory regimes for the same physical support. The factor b and the deficit r are actual acquisition masses, not a renormalized conditional experiment. The theorem is valid through the zero-risk threshold; it is not obtained by discarding the vanishing stratum.

Proposition 5.2 (Action-dependent exact zero threshold)**.** *For the regenerative action family and arbitrary positive scalar costs, $\Psi_m(N;c) = 0$ if and only if there are at most m points C and an action a with*

$$N \frac{\nu_a(C)}{c_a} \geq 1. \tag{5.4}$$

Thus mixtures of actions cannot improve the zero-distortion acquisition threshold, although they may improve a positive-distortion tradeoff.

Proof. At a zero minimum, $h_a\nu_a$ is supported on C, and

$$1 \leq \sum_a x_a \nu_a(C) \leq \left(\max_a \frac{\nu_a(C)}{c_a}\right) \sum_a c_a x_a \leq N \max_a \frac{\nu_a(C)}{c_a}.$$

One action achieves the finite maximum. Conversely repeatedly use such an action and accept only C. The expected cost is $c_a/\nu_a(C) \leq N$, and one live label and the m output labels implement it. Continuous parts contribute no mass to C and have not been counted as atoms. □

6. Two raw-kernel realizations of the general theorem

The first example has classical correlated reports and a polynomial test closure. The second has noncommuting quantum effects and action-dependent projective supports. Both verify (2.4) directly and hence instantiate the same Theorem 3.1; neither is used to prove it.

Theorem 6.1 (Controlled classical observation)**.** *Let the physical state space be $[-1/2, 1/2]^d$, and let each action have a centered compact probability ν_a on that box. Prepare $x_0 = 0$. For unknown fixed $\theta = (\theta_a)_a \in [-\eta, \eta]^{\mathcal{A}}$, $0 \leq \eta < 1$, define*

$$K_a^\theta(dx', dy \mid x) = \left(1 + \frac{4\theta_a}{d}\langle x, y\rangle\right) \delta_y(dx')\nu_a(dy). \tag{6.1}$$

After a stop, audit the coordinates independently with values in $\{-1, 0, 1\}$ and probabilities

$$\mathbb{P}(Z_i = \pm 1 \mid x) = \frac{x_i^2 + 1/2 \pm x_i}{2}, \qquad \mathbb{P}(Z_i = 0 \mid x) = \frac{1}{2} - x_i^2.$$

Then the task mean is $X_t = Y_t$, its conditional variance is $B = d/2$, and Theorem 3.1(iii) *holds uniformly in the unknown parameter with $\underline{b}_a = 1 - \eta$, $\overline{b}_a = 1 + \eta$. If all $\theta_a = 0$, its exact parts apply with the different laws ν_a. Every regular, singular or atomic choice of these acquired laws satisfies the corresponding conclusions whenever their separately stated geometric hypotheses hold.*

Proof. The likelihood is between $1 - \eta$ and $1 + \eta$, since $|\langle x, y\rangle| \leq d/4$. Centering of ν_a gives normalization. The actual first-call law from $x_0 = 0$ is ν_a, for every parameter. The report is the new state, and neither a rejected report nor the old state is reset to the preparation. The displayed row inequalities hold at every hidden state, hence at every conditional mixture induced by a history and any common policy. The audit probabilities are nonnegative on the box, sum to one, and give $\mathbb{E}Z_i = x_i$, $\mathbb{E}Z_i^2 = x_i^2 + 1/2$. This proves the mean and variance assertions.

These formulas arise from a finite affine executable-test representation rather than an assumed Fisher geometry. Let $\phi(x)$ be the vector of all monomials x^α with $\alpha_i \in \{0, 1, 2\}$, including one, and let its positive cone be generated by $\phi(x)$ on the physical box. The raw report instrument acts by

$$L_a^\theta(dy)z = \left(z_0 + \frac{4\theta_a}{d} \sum_i y_i z_{e_i}\right) \phi(y)\nu_a(dy).$$

It is positive on that cone and normalized on its base. The full protected audit probabilities are linear combinations of these monomials. Report updates, future actions and audit tests therefore lie in a genuine raw positive realization. No multiplication closure beyond this explicitly verified family is presumed.

For two incoming physical states the report laws have total variation at most $\eta\|x - x'\|/\sqrt{d}$ by the displayed affine density. The same-report current-task update is y for both states, so it has zero discrepancy. The global covers and small-ball constants, when required, are exactly those of the actually acquired ν_a. The program in the central theorem depends on these known measures, not on θ. □

Centering and a fixed affine rescaling embed the atomic–uniform law in Theorem 5.1 into this box; squared distortions acquire the square of the scale factor. One may take centered Cantor measures, smooth factors, or mixed dimensions for different actions. Their preparation and repeated acquisition costs remain those of (6.1).

Theorem 6.2 (Noncommuting action-dependent quantum supports)**.** *Let $q = 2^k$, $k \geq 1$, and for each action choose $r_a = 2^{j_a}$ with $1 \leq j_a \leq k$, an isometry $V_a : \mathbb{C}^{r_a} \to \mathbb{C}^q$, and a trace-preserving positive channel C_a with*

$$C_a(I_q/q) = I_{r_a}/r_a.$$

For example $C_a(\rho)$ may be a partial trace of a unitary conjugation on a factorization $\mathbb{C}^q = \mathbb{C}^{r_a} \otimes \mathbb{C}^{q/r_a}$. Set

$$T_a(\rho) = (1 - \eta_a)C_a(\rho) + \eta_a \operatorname{tr}(\rho) I_{r_a}/r_a, \qquad \eta_a \in [\eta_0, 1], \quad \eta_0 > 0,$$

and define the raw instrument, relative to Haar probability σ_{r_a} on projective pure states, by

$$L_a(d\psi)\rho = r_a \operatorname{tr}(P_\psi T_a(\rho))\, V_a P_\psi V_a^* \, \sigma_{r_a}(d\psi). \tag{6.2}$$

Prepare I_q/q. The channels and η_a may range over a known family with a fixed unknown member; the isometries and the current-state readout are known. Use the protected covariant audit on $\mathbb{C}^q$ with density $q \operatorname{tr}(\rho P_\zeta)$ and readout $Z = (q+1)P_\zeta - I_q$.

With Hilbert–Schmidt squared loss, the task mean is $X = V_a P_\psi V_a^*$, $B = q^2 + q - 2$, and the central envelope theorem applies with $\underline{b}_a = \eta_0$, $\overline{b}_a = r_a$. Its actual reference support has dimension $s_a = 2(r_a - 1)$ and satisfies

$$\nu_a\{P : \|P - P_0\|_{\mathrm{HS}} \leq u\} = (u^2/2)^{r_a - 1} \qquad (0 \leq u \leq \sqrt{2}). \tag{6.3}$$

The channels need not commute, and no common eigenbasis is used. When $\eta_a = 1$ for all actions the task is regenerative and the exact action profile applies.

Proof. The effect density in (6.2) is between η_0 and r_a on normalized positive inputs. Its integral is one because $\int P_\psi \, d\sigma_{r_a} = I_{r_a}/r_a$. The output normalized state is precisely the reported $V_a P_\psi V_a^*$, and the report law from the specified preparation is Haar for each action. Mixtures of unknown past states preserve the density bounds. These prove all conditional-envelope quantifiers, including histories after rejections.

For the audit, unitary invariance gives

$$\int P_\zeta \otimes P_\zeta \, d\sigma_q = \frac{I + \mathrm{Swap}}{q(q+1)}.$$

Indeed the integrand is supported on the symmetric subspace, its integral is scalar there by invariance, and its trace is one; the symmetric dimension is $q(q+1)/2$. Partial tracing the identity proves $\mathbb{E}[Z \mid \rho] = \rho$. Also $\|Z\|_{\mathrm{HS}}^2 = q^2 + q - 1$ for every audit outcome. Since every stopped state from (6.2) is pure, its conditional variance is $q^2 + q - 2$, independent of the action and parameter.

This is a raw finite-dimensional positive instrument on Hermitian matrices, with normalization tr. Future instruments and the audit are positive linear operations before normalization. No quantum system is available as free observer memory: only the classical labels in Section 2 are retained by

the observer. For normalized inputs ρ, ρ', report total variation is at most $\frac{1}{2}\|\rho - \rho'\|_1$, by positivity and trace preservation of the measurement channel. The current-task outputs agree for a common action and report. Covers and small-ball masses are obtained directly from (6.3). To verify that formula, choose $P_0 = e_1 e_1^*$; the random variable $|\psi_1|^2$ has density $(r_a - 1)(1-t)^{r_a-2}$, as follows by normalizing independent complex Gaussian coordinates. Since $\|P_\psi - P_0\|_{\mathrm{HS}}^2 = 2(1 - |\psi_1|^2)$, integration yields the displayed mass. These steps prove the same general theorem independently of the classical realization. □

A marked convex mixture of (6.2) with the positive operation $\rho \mapsto \mathrm{tr}(\rho)P_0$ produces an actual atom plus a continuous projective component. Choose P_0 outside the selected proper projective subspace when separation is desired. Both outputs remain pure, so the same audit variance applies. For $\eta_a = 1$ the mixed law is regenerative, and Theorem 5.1 gives its mass-dependent frontier. For $\eta_a < 1$ the conditional-envelope conclusions apply. This verifies a singular stratum in a noncommuting raw experiment; an arbitrary rank-changing filter is not being identified with it.

7. An exact informative Gaussian acquisition frontier

We now leave task regeneration. The current hidden state cannot be recovered from the last Gaussian report, and its full posterior depends on the past. The result below is a consequence of raw occupation balance and the classical likelihood-ratio principle [13]; it is not an application of (2.3).

Prepare $X_0 \in \{-1, 1\}$ uniformly. Before each report let the hidden chain change sign with probability $q \in (0, 1/2]$. Then report

$$Y_t \mid X_t = \pm 1 \;\sim\; N(\pm\mu, 1), \qquad \mu > 0,$$

independently conditional on the hidden path. The convention is *transition first, report second.* After the observer stops, the protected audit is X_τ, with no intervening transition. The observer has exactly one live state and one terminal state, total $M = 2$, and must use at least one call. Thus its most general Borel randomized program stops with a probability $u(y) \in [0, 1]$ depending only on the current report. Its terminal prediction is one constant. The parameters q, μ are known; this theorem is not an unknown-parameter minimax claim.

Put $\kappa = (1 - 2q)/(2q)$ and let $\overline{\Phi}$ be the standard normal survival function. For $h \in \mathbb{R}$ define

$$\begin{aligned} a_+(h) &= \overline{\Phi}(h - \mu), & a_-(h) &= \overline{\Phi}(h + \mu), \\ D(h) &= a_+(h) + a_-(h) + 2\kappa a_+(h)a_-(h), & & \\ T(h) &= \frac{2 + \kappa\{a_+(h) + a_-(h)\}}{D(h)}, & m(h) &= \frac{a_+(h) - a_-(h)}{D(h)}. \end{aligned} \tag{7.1}$$

Theorem 7.1 (All-program two-state frontier)**.** *For every $N > 1$ there is a unique $h_N \in \mathbb{R}$ with $T(h_N) = N$, and*

$$\mathcal{R}_{\mathrm{aut}}(N, 2) = 1 - m(h_N)^2. \tag{7.2}$$

An attaining program stops exactly when $Y_t \geq h_N$ and predicts $m(h_N)$. At $N = 1$ the optimum is one, attained by stopping after every first report and predicting zero. The lower bound covers all Borel randomized two-state programs, not only threshold candidates.

Proof. For an arbitrary legal routing u, let $a_\pm = \int u(y)\varphi(y \mp \mu)dy$. Positive equivalent report densities imply $a_- = 0$ if and only if $a_+ = 0$; that case never stops and is inadmissible. Let $v_\pm$ be the actual expected numbers of pre-decision reports in states ± 1, and let $d_\pm = a_\pm v_\pm$ be the probabilities of the two hidden states at stopping. Write $t = v_+ + v_- = \mathbb{E}\tau$ and $m = d_+ - d_-$. These are actual finite occupations, not a formal invariant solution. With

$$P = \begin{pmatrix} 1 - q & q \\ q & 1 - q \end{pmatrix}, \qquad v = \tfrac{1}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + P\,\mathrm{diag}(1 - a_+, 1 - a_-)v,$$

subtracting the two components gives $v_+ - v_- = -\kappa m$, whereas adding gives $a_+v_+ + a_-v_- = 1$. Solving these two equations and the definition of m yields

$$t = \frac{2 + \kappa(a_+ + a_-)}{a_+ + a_- + 2\kappa a_+ a_-}, \qquad m = \frac{a_+ - a_-}{a_+ + a_- + 2\kappa a_+ a_-}. \tag{7.3}$$

Conversely any nonzero routing has a uniform per-call stop probability at least $\min(a_+, a_-) > 0$, so these equations describe its actual occupation and finite mean. Its optimal constant prediction is m, with risk $1 - m^2$.

Reflecting $u(y)$ into $u(-y)$ if necessary preserves cost and risk and makes $a_+ \geq a_-$. At fixed $a_- = b > 0$, increasing $a_+ = a$ increases m and decreases t. For the common denominator $D = a + b + 2\kappa ab$, the relevant derivatives are

$$\frac{\partial m}{\partial a} = \frac{2b + 2\kappa b^2}{D^2} > 0, \qquad \frac{\partial t}{\partial a} = -\frac{2(1 + \kappa b)^2}{D^2} < 0.$$

The Gaussian likelihood ratio $\varphi(y - \mu)/\varphi(y + \mu) = e^{2\mu y}$ is strictly increasing. Among all $0 \leq u \leq 1$ with given a_-, the upper-tail rule maximizes a_+. One can see this without any existence theorem: choose h having the same negative-state tail mass, multiply $u_h - u$ by $\varphi(y - \mu) - e^{2\mu h}\varphi(y + \mu)$, and integrate. The integrand is nonnegative and the second term cancels. Thus every randomized routing is dominated in both cost and risk by a threshold routing.

For thresholds, $T(h)$ is continuous and strictly increasing from one to infinity. Monotonicity follows also pathwise from nested threshold hitting times, with strictness from the positive probability that the first report lies between two distinct thresholds. The limit at infinity follows from $a_\pm \to 0$. The ratio $R(h) = a_+(h)/a_-(h)$ is strictly increasing: it is the conditional mean of the strictly increasing likelihood ratio under the negative-state tail law. Since $a_+(h)$ decreases and

$$m(h) = \frac{R(h) - 1}{R(h) + 1 + 2\kappa a_+(h)},$$

$m(h)$ strictly increases from zero to one. A feasible threshold with mean less than N can therefore be raised to h_N, improving its risk. Combining this fact with the preceding domination proves the all-program lower and the attaining construction. At $N = 1$, $\tau \geq 1$ and $\mathbb{E}\tau \leq 1$ force a stop after one call almost surely; equivalent report densities force $u = 1$ almost everywhere, so its constant Bayes prediction is zero. □

Proposition 7.2 (A same-task checkpoint comparison)**.** *Fix the optimal threshold $h = h_N$ and the same physical stopping protocol. Let $h = t_0 < t_1 < \cdots < t_m = +\infty$ be fixed report-bin boundaries. Put*

$$v_\pm = \frac{N \mp \kappa m(h)}{2}, \quad p_{j,\pm} = v_\pm\{\overline{\Phi}(t_{j-1} \mp \mu) - \overline{\Phi}(t_j \mp \mu)\}.$$

A checkpoint encoding only the final report bin, with m fixed decoder values, has exactly

$$R_{\mathrm{cp,bin}} = 1 - \sum_{j=1}^{m} \frac{(p_{j,+} - p_{j,-})^2}{p_{j,+} + p_{j,-}}. \tag{7.4}$$

Terms of zero mass are zero. In particular the unrestricted-history optimum obeys $\mathcal{R}_{\mathrm{cp}}(N, m) \leq R_{\mathrm{cp,bin}}$. This is an attainable checkpoint upper bound, not an equality with the unrestricted optimum.

Proof. Sum the raw report density over the actual pre-report occupation. The terminal joint density of (X_τ, Y_τ) is $v_\pm\varphi(y \mp \mu)\mathbf{1}_{y \geq h}$. Integrating each bin gives $p_{j,\pm}$. Their total mass is one by the stop equation. The conditional mean on a bin is $(p_{j,+} - p_{j,-})/(p_{j,+} + p_{j,-})$; conditional squared projection gives (7.4). The checkpoint may use the entire history but is allowed to use only its final report, so the inequality follows. Different unrestricted stopping protocols may improve it; no optimality for that larger class is inferred. □

Certified finite-parameter benchmark. The accompanying arithmetic certificate specializes to $q = 1/4$, $\mu = 1$ and budgets $N = 2, 4, 8$. It encloses h_N by rational bisection, evaluates (7.2) with outward interval bounds, and applies (7.4) with two and four fixed report bins. Normal tails are bounded by an alternating power series with an explicit remainder and rational enclosures of π and its square root; no empirical tail truncation is used. The native source includes its generated table below. These checks validate finite arithmetic and an executable example; the continuum all-program proof is the preceding theorem.

N	h_N	$R_{\mathrm{aut}}(N, 2)$	checkpoint, $m = 2$	checkpoint, $m = 4$
2	$[-0.289093, -0.289092]$	$[0.821792, 0.821793]$	$[0.653114, 0.653115]$	$[0.619382, 0.619383]$
4	$[0.806408, 0.806409]$	$[0.312884, 0.312885]$	$[0.303481, 0.303482]$	$[0.297684, 0.297685]$
8	$[1.589239, 1.589240]$	$[0.084299, 0.084300]$	$[0.083987, 0.083988]$	$[0.083669, 0.083670]$

The checkpoint columns are outward enclosures of achievable risks using the feasible lower threshold, not certified intervals for the checkpoint optimum. Their two-bin boundary is $h + 1$; the four-bin interior boundaries are $h + 1/2, h + 1, h + 2$. Both tails are retained.

The threshold is a program constant, not a persistent real-valued state. A rational lower approximation to h_N has mean at most N and approaches the optimal risk continuously. A bounded current-report arithmetic routine needs its own workspace and accuracy certificate; Section B describes the resource distinction. The Gaussian theorem does not imply a multi-state or unrestricted-checkpoint closed form, nor does it solve the separate fixed-horizon uniform-constant benchmark requested for the older filtering theorem.

8. Common-task error composition and physical resources

Proposition 8.1 (Typed causal simulation)**.** *Suppose a parameter-independent causal simulator takes a target experiment into a source experiment. Its report transformation, strategy lift, available actions, stopping decision and protected task readout are compatible. Suppose it uses at most R internal states, including its clock, has a verified mean native-call bound $\kappa N + c$ for every simulated target procedure of mean at most N, and approximates the complete stopped and scored target law in total variation by at most ε. For a common loss in $[0, L]$,*

$$\mathcal{R}_{\mathrm{source}}(\kappa N + c, RM) \le \mathcal{R}_{\mathrm{target}}(N, M) + L\varepsilon. \tag{8.1}$$

Two such simulations compose with additive deficiencies, state multiplier R_1R_2, and mean-call bound $\kappa_1\kappa_2 N + \kappa_1 c_2 + c_1$. Program descriptions, temporary workspace and physical execution time are separately charged.

Proof. Run the target observer and simulator in their product state, of size at most RM. The lifted actions and the joint clock are causal because only past reports and fresh coins are used. The assumed mean certificate applies to that executed policy. The total variation convention is $\|P - Q\|_{\mathrm{TV}} = \sup_A |P(A) - Q(A)|$, under which the difference in expectations of a loss in $[0, L]$ is at most $L\varepsilon$. Infimizing gives (8.1). Composition follows from substitution of the mean bound, the product of states, and kernel contraction and the triangle inequality for total variation. A reverse risk bound needs a reverse simulator or a separate converse; it does not follow by reversing this inequality. □

If implementation errors can be coupled at a common pre-call history with mismatch probability at most $\eta(h)$, and the final protected audit with error $\eta_f(h)$, first-discrepancy coupling gives the complete-task error bound

$$\min\left\{1, \mathbb{E}\sum_{t=1}^{\tau} \eta(\mathcal{F}_{t-1}) + \mathbb{E}\eta_f(\mathcal{F}_\tau)\right\}. \tag{8.2}$$

Indeed the first mismatch at call t has probability at most the ideal expectation of $\mathbf{1}_{\{\tau \ge t\}}\eta(\mathcal{F}_{t-1})$; sum these events and then the protected audit event. A nonterminating approximate trajectory

is recorded as a failure outcome. This proves a bounded-risk comparison, not an expected-cost comparison. Theorem 3.1(iv) supplies the additional same-budget repair for regenerative tasks. The general stopped perturbation theorem in the companion [1] treats nonregenerative raw instruments with a different, square-root, modulus.

For a fixed stopped law, a decoder error of norm at most v obeys

$$\sqrt{\mathbb{E}\|X_\tau - \widehat{r}_J\|^2} \le \sqrt{\mathbb{E}\|X_\tau - r_J\|^2} + v. \tag{8.3}$$

This is the L^2 triangle inequality. It does not apply unchanged when numerical error also changes the stopping law; that effect must first be controlled by (8.2) and an appropriate budget certificate. A raw likelihood error, a numerical coordinate error, and an unobserved calibration shift are distinct quantities.

Theorem 8.2 (Attained joint law through a singular threshold)**.** *Take the regenerative atomic–regular experiment of Theorem 5.1. On accepting a core report, freeze the core state and perform exactly one additional bit acquisition. An unknown fixed sign $\chi \in \{-1, 1\}$ is reported correctly with probability $1 - e$ and is erased with probability e. It affects no preceding core report. Another unknown fixed sign $\omega \in \{-\delta, \delta\}$ affects no observer report. The observer commits predictions of the core audit, χ, and ω, and one protected joint audit then reveals their scored values. No core restart or bit reread is legal.*

For $0 \le e \le 1$, $0 \le \delta \le 1$, $n \ge 1$ and $M \ge 9$, put $\varepsilon = e/2$ and $r = [1 - n(1 - b)]_+$. Then

$$\mathcal{R}_{\mathrm{ext}}(n, M, \delta, \varepsilon) - B \asymp \frac{r^{1+2/s}}{(nb)^{2/s} M^{2/s}} + \delta^2 + \varepsilon. \tag{8.4}$$

The constants are uniform in these budgets and error parameters and depend only on the regular-support data. The bit source has causal deficiency exactly ε relative to the exact bit report with the same protected truth. An attaining-order implementation uses $1 + 4m \le M$ total states, where $m = \lfloor (M - 1)/4 \rfloor$.

Counting one initial preparation, all core acquisitions, the bit call and the final audit gives total physical mean budget $\mathsf{N} = n + 3$. Thus in total-call notation $n = \mathsf{N} - 3$ and $\mathsf{N} \ge 4$; these three calls are not hidden in the geometric term.

Proof. Use the one-live-state core construction with m centers. Its m accepted labels are bit-waiting states, not final outputs. The bit has three reports, leading to $3m$ terminal labels whose bit decoders are $+1, 0, -1$. The calibration decoder is zero. Total memory is one core-live state, m bit-waiting states and $3m$ final labels. The geometric upper from Theorem 5.1 uses $m - 1$ continuous centers; for $M \ge 9$, $m - 1 \ge M/16$. Add the exact bit squared risk e and calibration risk δ^2 to get the upper, including the region $r = 0$.

For a converse, any legal extended procedure chooses its core stop before the bit report. The core law therefore has continuous terminal mass at least r and density cap nb. The first component of its eventual decoder has at most M values. The lower proof of Theorem 5.1, applied to that full codebook, still holds even if a later bit report selects among those values. Average over the two signs χ and ω. Conditional on erasure, all pre-audit information is independent of χ, so the squared bit risk is at least one on that event. The calibration sign is independent of all such information, and its averaged squared risk is at least δ^2. These losses add to the core lower. A maximum over the four sign alternatives is at least their average. This proves (8.4).

On a nonerased bit, the simulator outputs the report; on erasure it guesses a fair sign. It is parameter independent, preserves the protected truth and has error $e/2$ for each sign. Conversely under a uniform prior on the two signs no pre-audit simulator has information about the sign on erasure, so its average error is at least $e/2$. The worst sign error is at least that average. This proves exact deficiency. It is an attained interface defect, not merely an allowed tolerance. The state and physical-call counts follow from the stated protocol. □

The hidden-offset and erased-bit devices are inherited from the earlier same-task constructions [1, 2]. What changes in (8.4) is their conjunction with a mass-dependent singular acquisition frontier. A permission to make error δ or ε need not force that much risk in an arbitrary experiment.

Appendix A. Predictive realization and the inherited general mechanism

This appendix states the precise connection to the mother experiment. It prevents a current task vector from being confused with the minimal state for every executable future test.

Let $\mathcal{H}$ be the standard Borel disjoint union of finite visible histories, with available actions, stop permissions and all interface coordinates recorded. Choose a countable determining family (g_j) of bounded future tests, containing the required report cylinders and closed under each legal one-step continuation. Relative to a specified prepared law, write

$$q(h) = (\mathbb{E}[g_j \mid h])_{j\geq 1}$$

and adjoin the interface coordinates. Countable determination means that equality of these values is equivalent to equality for the entire declared test family. Conditional test values are given jointly Borel versions. Assume explicitly that $q(\mathcal{H})$ is Borel and that q has a Borel right inverse on its image. A general Borel surjection is *not* silently assumed to have such a right inverse.

Proposition A.1 (Realization and minimality)**.** *Under these hypotheses, predictive equivalence $h \sim h'$ if and only if $q(h) = q(h')$ has a standard Borel realization. Its legal report laws and its next predictive state admit jointly Borel versions depending on $(q(h), a, y)$. They agree with the history kernels at almost every report under each actual row. If another statistic $z(h)$ is sufficient for all declared future tests, in the sense that every test value is a Borel function of $z(h)$, then q is a Borel function of z. Thus the realization is minimal for those tests and that preparation.*

Proof. The image is standard Borel by assumption and its fibers are exactly the equivalence classes. At state u choose the history $\sigma(u)$ given by the Borel right inverse, and define the report and next-state kernels from that history. These are jointly Borel. One-step test closure implies that histories in a fiber have the same integrals of every report cylinder times every next test value. Their report laws are equal, and uniqueness of conditional expectations implies equality of each next-test coordinate almost everywhere under that common law. Countability lets us take a common null set for the coordinates. Hence the next predictive state agrees with the actual history update there. Off that set use the chosen Borel version; no positive-probability continuous singleton is used. Finally the coordinate functions through z assemble into a Borel map into the countable product, with values in the declared image on actual histories, proving minimality. □

The finite raw realizations in Section 6, with their model fixed and known, give such a representation directly: the reported classical state or pure density matrix determines all future kernel words and is separated by the audit means. No abstract selection assumption is needed there. If the model is unknown and retained under a prior, its posterior may be part of the full predictive state. The current known task response can still be report-recoverable and satisfy (2.4); our proof does not assume that it alone summarizes all future parameter information. Likewise the Gaussian posterior is a history-dependent state, not its current report.

General common-program occupation. For orientation we record the mechanism retained with complete proofs in [1]. A finite ordered raw carrier (E, C, e) has positive report maps $L_a(y)$ with $\int eL_a(y)x\, d\lambda_a = ex$. A homogeneous live-state program induces a positive substochastic map P on the product carrier and an exit map Q. With prepared vector a, its actual live occupation and exit mass are

$$z_{\mathrm{act}} = \sum_{t\geq 0} P^t a, \qquad d = Qz_{\mathrm{act}}. \tag{A.1}$$

A finite candidate satisfying $z = a + Pz$ need not itself be acquired: iteration gives

$$z = \sum_{t=0}^{n-1} P^t a + P^n z.$$

If the exit mass is one, the decreasing positive limit $h = \lim_n P^n z$ is invariant, $Qh = 0$, and $z_{\mathrm{act}} = z - h \le z$. Thus nonnegative upper-cost constraints survive removal of inaccessible circulation, but that circulation may not be counted as observed mass. The central regenerative profile is the case in which this actual-action constraint can be solved explicitly, preserving an entire terminal distribution with one live label. The Gaussian proof solves (A.1) for a genuinely informative two-state raw carrier.

For an unknown fixed model, a *single* routing kernel must solve the balance constraints in every model. The retained common-kernel separation theorem sums model and state contributions before maximizing over output labels. Exchanging those operations would permit model-dependent programs. The exact action profile in the present paper has no hidden use of that invalid exchange; its unknown-model extension is instead the explicit common program in Theorem 3.1(iii).

Informative Bayesian compression is not discarded. The likelihood-weighted identity from [2] also remains in force. Let affine raw functions $U(y, x)$ and $\ell(y, x) \ge 0$ describe a finite future-test family, and let $F = U/\ell$ on positive evidence. For an actual incoming predictive vector X and retained label S, put $V = \mathbb{E}[X \mid S]$ and $W = F(Y, V)$. Affine integration before normalization gives $W = \mathbb{E}[X' \mid S, Y]$. For any fresh randomized outgoing label J, define $c_{sj} = \mathbb{E}[W \mid S = s, J = j]$ and $v_j = \mathbb{E}[W \mid J = j]$. Two conditional projections yield

$$\mathbb{E}\|X' - v_J\|^2 = \mathbb{E}\|X' - W\|^2 + \mathbb{E}\|W - c_{SJ}\|^2 + \mathbb{E}\|c_{SJ} - v_J\|^2. \tag{A.2}$$

Each term is nonnegative. The first is equivalently the Jensen defect of $\int \|U(y, x)\|^2/\ell(y, x)\, d\lambda(y)$. Zero-evidence reports contribute zero. The exact identity alone does not supply a globally compatible compressed recursion. The prior global-cover and curvature or same-report stability theorems supply sufficient hypotheses for that different task, with their complete proofs retained. Neither task regeneration nor one-step centroid equality is substituted for those hypotheses.

Example A.2 (Local exact cells do not supply lost continuation information)**.** Prepare two independent fair signs U, V and reveal both at the first call. At the second call an independent uniform query $Q \in \{1, 2\}$ is reported; the protected target is $Z = U$ if $Q = 1$ and $Z = V$ otherwise. If the cut retains only $S = U$, then after the query

$$W = \mathbb{E}[Z \mid S, Q] = S\mathbf{1}_{\{Q=1\}}.$$

Three final labels encode $W \in \{-1, 0, 1\}$ exactly. Every row quantization defect and every pooled-centroid defect in (A.2) is zero, but the incoming-information defect is $\mathbb{E}(Z - W)^2 = 1/2$. More generally, a zero-error answer for both future queries requires at least four cut labels. For randomized encoding, the supports of the cut-label distributions of two different sign pairs must be disjoint: the pairs differ in a coordinate, and that query has positive probability, so a common label would have to decode both signs without access to the discarded pair. Hence four disjoint nonempty supports are necessary. The example concerns the cut, not a free-clock total-state formula. It explains why local centroid equalities cannot replace a global acquired-occupation or continuation certificate.

Appendix B. Finite descriptions, arithmetic and implementation

The state-optimal results concern Borel randomized observers. They do not identify their persistent-label count with their program length or temporary computational space. We give a precise effective qualification for the regenerative class.

Assume that H has computable finite h-nets, and that each response law is supplied with finite Borel partitions of cell diameter at most h, with interval-computable cell masses. A partition-membership primitive for the current physical report is supplied together with its program, workspace, time and error certificates. This last input is substantive: arbitrary computable measures do not automatically have computable masses on every chosen boundary. The laws, nets, partitions, finite action set, unit costs and real budget N are known. For numerical search N is computable, and a positive margin from one is given when $N > 1$.

Proposition B.1 (Same-state finite-precision implementation)**.** *Under these effective inputs, for every $N \geq 1$, $m \geq 1$ and $\alpha > 0$ there is a finitely described one-live-state observer with m terminal labels, mean acquisition at most N, and risk at most $B + \Psi_m(N) + \alpha$. Its coefficients and decoders can be chosen rational. If there are at most K_h cells per action, an explicit table description uses*

$$O\big(|\mathcal{A}|K_h(m+1)\log Q + mdp + |\mathcal{A}|\log Q\big) \tag{B.1}$$

program bits, in addition to the given partition primitive, for routing denominator Q and p-bit decoder coordinates. Temporary workspace for evaluating and sampling a row is

$$O\big(\log(|\mathcal{A}|K_h(m+1)) + \log Q + dp\big) \tag{B.2}$$

plus the primitive workspace. This is an existence and exhaustive-synthesis statement, not an efficient-planning or optimal-workspace theorem.

Proof. Start with an optimal one-live-state Borel program. Replace, for each action and partition cell, its terminal-label probabilities by their cell averages under ν_a. This preserves every action-indexed terminal-label mass and the one-call stopping probability $s \geq 1/N$. Within a cell, squared distances to a fixed decoder vary by at most $2Dh + h^2$, where D bounds the diameter of the task domain. Replacing decoder values by an h-net adds at most the same order. Summing over accepted mass and dividing by s shows that the risk increases by at most $4Dh + 2h^2$, not by a factor proportional to an uncharged horizon. We may approximate the finitely many averages by rational numbers.

To keep the budget exact during rational approximation, first redirect a fraction ρ of continuations to one existing terminal label. If $N > 1$, the resulting stopping mass satisfies

$$s_\rho - 1/N \geq \rho(1 - 1/N) > 0,$$

and its excess-risk increase is at most $\rho L(N-1)$. Rounding the action and row simplices to denominator Q changes the one-call joint law by at most $d_Q = (|\mathcal{A}| + m + 1)/Q$ up to a fixed harmless factor that can be doubled in choosing Q. Choose Q so that this defect and the cell-mass evaluation error are less than one quarter of the displayed stopping margin. Then the stopping probability has a certified lower bound $1/N$, hence the mean stays at most N. Its risk perturbation is bounded by $2LNd_Q$ after increasing the harmless rounding constant if necessary. Choose h, ρ, Q and the mass precision successively so the sum of these errors is below α.

At $N = 1$ the exact program accepts on every call. Keep the continuation probabilities exactly zero, and round only action and terminal-label probabilities; stopping remains certain. The same cell and decoder estimates converge to zero. Exhaustively enumerate the finite rational tables and center grids, keeping only those with a certified stop probability at least $1/N$. The preceding construction proves that this search contains an α-optimal table once its parameters are fine enough. Risk evaluation is a finite sum of interval-computable masses and bounded coefficients. This argument does not assume an effective algorithm for finding a Borel optimizer before enumeration.

Storing all table entries and decoder coordinates gives (B.1). During execution only the row index, current random integer and current coordinate calculations need workspace, giving (B.2); no such value persists across calls. Rational sampling may use a finite-denominator uniform-integer primitive, or fair-bit rejection sampling with its own expected random-bit and time cost. Exact fair-bit implementation does not give a deterministic worst-case random-bit bound for arbitrary denominators. The expected number of table executions is at most N; multiply the declared

per-execution primitive cost accordingly. Offline enumeration and integration costs remain separate and may be very large. □

The term "rational decoder" in the proposition allows a rational approximation in a fixed ambient box; if H has no suitable rational points, project conceptually for the comparison but execute the nearby rational box decoder. Squared-loss continuity gives the same estimate with a slightly enlarged fixed diameter. Neither this observation nor the table count supplies a free high-precision physical report. Nonzero classification-error probabilities must enter the complete-task coupling and budget repair separately.

The normal-tail certificate used in Section 7 is elementary. For rational $x \geq 0$,

$$\Phi(x) = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{2^k k!(2k+1)}.$$

After an index for which successive absolute terms decrease, the tail lies between the next two alternating partial sums. Rational bounds on π follow from Machin's identity and the alternating arctangent series; rational bisection of a square gives bounds on $(2\pi)^{-1/2}$. Symmetry treats negative arguments. The implementation uses these intervals, not floating-point values of an unproved external special-function approximation. The bisection endpoints themselves give a feasible lower-threshold digital program and a rigorous bracket on its continuum optimum. This arithmetic is a verification of the displayed example, not a proof of optimality over a continuum of routing functions.

Appendix C. Two distinct stability mechanisms

The curvature and same-report alternatives in the informative Bayesian theorem [2] are genuinely different. The examples below address that comparison directly. They do not assert that a stability certificate alone proves the additional geometric hypotheses.

For a homogeneous affine Bayesian experiment on a fixed compact convex domain K, write

$$\Phi(u) = \int \|F(y,u)\|^2 \ell(y,u)\, d\lambda(y).$$

The curvature route asks for $\frac{1}{2}D^2\Phi(u)[v,v] \leq b_t\|v\|^2$ with products of the b_t decaying uniformly. The same-report route asks for uniformly bounded metrics dominating the task norm and deterministic bounds on $\mathrm{dist}_t(F(y,u), F(y,v))$ whose products decay uniformly for all common nonexceptional reports. The latter is not merely an average contraction condition.

Proposition C.1 (Neither stability route implies the other)**.** *There are explicit positive prepared finite-state experiments with the following properties.*

(a) *A two-state Gaussian experiment satisfies the same-report route on its full invariant prediction domain, but every valid uniform Euclidean curvature factor is greater than* $48020000/46268279 > 1$.

(b) *A two-state hold–reset experiment has curvature factor a for any chosen $0 < a < 1$, but cannot satisfy the same-report route with uniformly bounded, norm-dominating metrics. This second example separates stability certificates; its atomic row laws do not satisfy the positive-dimensional small-ball premise of the older regular-geometry theorem.*

Proof. For (a), let $q = 1/100$, $\lambda = 1 - 2q = 49/50$, and

$$P = \begin{pmatrix} 1-q & q \\ q & 1-q \end{pmatrix}, \quad f_\pm(y) = \varphi(y \mp 1/2), \quad L_y = P\,\mathrm{diag}(f_+(y), f_-(y)).$$

This example uses *report first, transition second*, exactly as in the older informative filtering certificate; it is separate from Section 7. The invariant domain is $K = P\Delta_1$, represented by $p \in [q, 1-q]$. Its categorical forecast is $(p, 1-p)$. Gaussian densities are positive and every interior point of K is in the support of every acquired posterior row.

In the log-odds metric $d(p,p') = |\log(p/(1-p)) - \log(p'/(1-p'))|$, a Bayesian report adds the same log likelihood ratio and is an isometry. The transition $p \mapsto q + \lambda p$ has derivative in log-odds coordinates

$$\frac{\lambda p(1-p)}{(q+\lambda p)(1-q-\lambda p)} \le \lambda < 1,$$

since the denominator minus $p(1-p)$ equals $q(1-q)(2p-1)^2 \ge 0$. Thus the full normalized update contracts by λ. On K this metric has finite diameter and is uniformly equivalent to the task norm. It supplies the same-report route.

For the curvature route, direct differentiation of the raw perspective gives, relative to the categorical tangent norm $\|(v,-v)\|^2 = 2v^2$, the exact factor

$$\beta(p) = \lambda^2 \int \frac{f_+(y)^2 f_-(y)^2}{\{p f_+(y) + (1-p) f_-(y)\}^3}\, dy.$$

On a compact interior neighborhood of the point used next the integrands and derivatives are bounded by a constant times $f_+ + f_-$, justifying differentiation under the integral. Since $f_+/f_- = e^y$ and $f_+^2/f_- = e\,\varphi(y-3/2)$, restriction to $y \le 2$ gives

$$\beta(p) \ge \frac{\lambda^2 e\,\Phi(1/2)}{\{1+p(e^2-1)\}^3} > \frac{\lambda^2 (27/10)(1/2)}{(1+7p)^3}.$$

Here $e > 27/10$, $e^2 < 8$ follow directly from the exponential series, and $\Phi(1/2) > 1/2$ by symmetry. The point $p = 11/1000$ is strictly inside K. Substitution gives $48020000/46268279 > 1$. Therefore every homogeneous-domain curvature factor must exceed this number at every phase, and its products cannot decay. This is a failed sufficient curvature certificate, not a failed filtering theorem.

For (b), take hidden categorical states with prediction $u \in \Delta_1$, prepare the known law $u_0 = (3/4, 1/4)^T$, and use two report symbols H, R with raw instruments

$$L_H = aI, \qquad L_R = (1-a)\pi \mathbf{1}^T, \qquad \pi = \tfrac{1}{2}\begin{pmatrix}1\\1\end{pmatrix}.$$

Report H keeps the hidden state, while report R resets it to π. Both probabilities are positive and independent of the input. The normalized updates are $F_H(u) = u$ and $F_R(u) = \pi$, so

$$\Phi(u) = a\|u\|^2 + (1-a)\|\pi\|^2, \qquad \tfrac{1}{2} D^2\Phi(u)[v,v] = a\|v\|^2.$$

Thus the curvature route holds with geometric factor $a < 1$. At every positive time the actually acquired prediction law assigns positive mass to both u_0 (no reset yet) and π (at least one reset). The counterexample therefore persists on their convex hull, and does not rely on states that the preparation never acquires.

If the same-report route held, compose its inequalities along the report word H^n, which has positive probability for every n. For any two distinct u, v in the declared domain this would give

$$\mathrm{dist}_n(u,v) \le C_0 \rho^n \,\mathrm{dist}_0(u,v).$$

Uniform domination of $\|u-v\|$ by the left side and bounded initial diameter contradict this as $n \to \infty$. A zero-probability exception cannot remove the positive-mass symbol H. This proves the asserted separation. □

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